

Andrii Skliar

STAFF DEEP LEARNING ENGINEER @ NVIDIA

Amsterdam, Netherlands • Citizenship: Netherlands

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Personal Profile

AI Research Engineer with 5+ years of experience building and scaling ML systems, with a focus on efficient inference, model compression, and real-world deployment. Co-inventor on multiple granted patents; contributed to ML research adopted in commercial products. Experienced in bridging foundational research and production-ready solutions in LLMs, NAS, and evaluation infrastructure.

Tools & Platforms: Python, PyTorch, Ray, Kubernetes, Docker, GCP, Git

ML Focus Areas: LLMs, MoEs, NAS, Quantization, Inference Optimization, Synthetic Data Generation

Work Experience

Nvidia

Amsterdam, Netherlands

Staff Deep Learning Engineer

Oct 2025 – Present

- Drive high-performance LLM inference on new NVIDIA GPU architectures through work spanning vLLM, SGLang, and low-precision serving.
- Design LLM architectures for both training and inference efficiency, with a focus on Mixture-of-Experts, serving-aware design, and deployable large-scale systems.
- Own optimization of attention, KV-cache, quantization, compression, and MoE inference paths, improving throughput, latency, and memory efficiency in production serving stacks.

Contextual AI

San Francisco, CA (Remote)

Member of Technical Staff (Applied Research)

Jan 2025 – Oct 2025

- Built and scaled synthetic data generation pipelines supporting 10+ enterprise LLM applications, enabling rapid dataset creation for evaluation, fine-tuning, and retrieval-augmented generation systems.
- Led end-to-end development of production ML systems, from data generation and model training to deployment and monitoring, improving model quality by 20% and increasing production reliability to 90%.
- Designed evaluation frameworks for instruction-following, grounding, and technical reasoning, establishing standardized benchmarks for enterprise LLM deployments.
- Collaborated with product and infrastructure teams to integrate model improvements into customer-facing systems and large-scale inference pipelines.

Qualcomm AI Research

Amsterdam, Netherlands

Senior Research Engineer

Oct 2019 – Jan 2025

- Led hardware-aware ML research under Prof. Max Welling, developing efficient model architectures and compression techniques including DONNA (hardware-aware NAS), simulated quantization, and sparse training methods.
- Contributed to the AIMET model optimization toolkit, enabling quantization-aware training and deployment of efficient neural networks on mobile and edge hardware.
- Demonstrated simulated quantization techniques that achieved up to 17% real-world energy savings in deployed neural networks.
- Built Qualcomm's first hybrid LLM architecture, enabling efficient on-device mixture-of-experts decoding and integration into production systems.
- Co-inventor on 7 US patents (5 granted, 2 pending) related to neural network efficiency, quantization, and hardware-aware model optimization.

QUVA Lab

Amsterdam, Netherlands

Research Intern

May 2019 – Oct 2019

- Master's thesis internship in collaboration with QUVA Lab under the supervision of Maurice Weiler.
- Extended the theory of Hyperbolic Neural Networks to Convolutional and Graph Neural Networks.

Selected Publications

Efficient LLM Inference using Dynamic Input Pruning and Cache-Aware Masking (**Patent Pending**)

Proceedings of Machine Learning and Systems (MLSys), 2025

Mixture of Cache-Conditional Experts for Efficient Mobile Device Inference (**Patent Pending**)

Transactions on Machine Learning Research, 2025

Think Big, Generate Quick: LLM-to-SLM for Fast Autoregressive Decoding (**Patented under Qualcomm IP portfolio**)

Proceedings of the 2nd Workshop on Efficient Systems for Foundation Models (ES-FoMo II) at ICML, 2024

Hyperbolic Convolutional Neural Networks

Arxiv Preprint (first author), 2023

Cyclical pruning for sparse neural networks (**Patented under Qualcomm IP portfolio**)

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022

Revisiting single-gated Mixtures of Experts (**Patented under Qualcomm IP portfolio**)

Proceedings of the 33rd British Machine Vision Conference (BMVC), 2022

Simple and Efficient Architectures for Semantic Segmentation

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022

Simulated Quantization, Real Power Savings (**Patented under Qualcomm IP portfolio**)

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022

Distilling optimal neural networks: Rapid search in diverse spaces (**Patented under Qualcomm IP portfolio**)

Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021

Adding object detection skills to visual dialogue agents

Proceedings of the European Conference on Computer Vision (ECCV) Workshops, 2018

Education

University of Amsterdam

Amsterdam, Netherlands

MSc in Artificial Intelligence

Sep 2017 - Aug 2019

- GPA: 8.8/10, Cum Laude
- Honours programme with multiple successful research projects.
- Courses in Theoretical and Applied Machine Learning with main focus on Advanced Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning and Information Theory.

NTUU "Kyiv Polytechnical University" ESC "Insitute of Applied System Analysis"

Kyiv, Ukraine

BSc in System Analysis

Sep 2013 - Jul 2017

- GPA: 4.74/5, Top 5% of the Year Group
- Focused on applied mathematics, statistics, and computer science foundations.